

**Name:**

Instead of a lab, today we have structured notes for today's lecture.

## Warm-Up: Prior Knowledge (Review from Last Week)

Fill in definitions for the terms we learned last week:

- Population:
  
- Population parameter:
  
- Sample:
  
- Sample statistic:

### Key idea:

In statistical inference we do not have data for the entire population of interest, so we use:

a \_\_\_\_\_ as an estimate for a \_\_\_\_\_.

## Today's Goal

We will understand how **sample statistics vary** from sample to sample, and why that matters for inference.

Today we will have data for our entire population of interest. We will draw a samples from this population to study how reliable our estimation is, and how to increase precision of our estimate.

By the end of class, you should be able to:

- Explain why sample statistics change
- Describe a **sampling distribution**

## Why Do We Sample?

Real-world data (identify these for the example from class slide 5):

- Population:
- Population parameter:

**Think:**

(*After slide 7*) Why don't we just always use the entire population?

## Sampling Variability

(*After slide 12*) Fill in:

- If we were to collect **multiple samples** of size  $n = 40$ , the statistic will:
- This happens because:
- We can use simulations to study the \_\_\_\_\_, this provides us with an understanding of how accurate and precise the sample statistic is as an estimate for the population parameter.

## Sampling distribution for a mean

The mean salary was actually \_\_\_\_\_ for all full time LAPD workers in 2018, and with our first sample of 50 workers we would have estimated it to be \_\_\_\_\_.

## What Does the sampling distribution Look Like?

We simulated 20,000 separate samples under different conditions, different sample sizes ( $n$ ). Describe the features of the sampling distributions.

- Shape (symmetric or skewed?):
- Center:
- Spread depends on:

## Summarizing

In this lecture we learned that a \_\_\_\_\_ is a good estimate for a population proportion. To increase the precision (lower the standard deviation) of our estimate we want sample with **more** / **fewer** observations (circle one).

In this lecture we learned that a \_\_\_\_\_ is a good estimate for a population mean. To increase the precision (lower the standard deviation) of our estimate we want sample with **more** / **fewer** observations (circle one).

## Bridge to Bootstrapping (week 13)

In real life we usually have:

- ☐ Data from the entire population of interest
- ☐ Many samples
- ☐ Only one sample

**Knowledge Check (Preparation for Lecture Quiz)**

1. What is the difference between a population parameter and a sample statistic?
2. Why can different samples from the same population produce different sample proportions (or different sample means)?
3. What happens when sample size increases?
4. What is a sampling distribution?
5. Why is the sampling distribution helpful?